



Individual Coursework Submission Form

Specialist Masters Programme

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MSc Actuarial Science 2025-26

SMM636 Machine Learning - Individual coursework

Heart Disease Classification Using Statistical and Machine Learning Methods

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1. Introduction

This report analyses the **heart-disease** dataset for **462 male patients** from a high-risk region of the Western Cape, South Africa. The response variable is **coronary heart disease (chd)**, coded as **1 = disease present** and **0 = disease absent**. The objective is to use these nine predictors to model and predict CHD and to evaluate the performance of several classification methods; systolic blood pressure (**sbp**), cumulative tobacco use (**tobacco**), low density lipoprotein cholesterol (**ldl**), **adiposity**, family history of heart disease (**famhist**), type-A behaviour (**typea**), **obesity**, alcohol consumption (**alcohol**), and **age**. Machine learning methods are used to improve predictive accuracy beyond traditional statistical models.

2. Exploratory Data Analysis

2.1 Data structure and missing values

The dataset contains **462 observations** and **10 variables** in total, including the binary outcome **chd**. The absence of missing data ensures that model estimates are not affected by imputation bias and allows all observations to be retained for analysis.

2.2 Class distribution

The outcome is moderately imbalanced. Out of 462 patients, **302 (65.4%)** do not have CHD and **160 (34.6%)** do have CHD. This is shown in **Figure 1**.

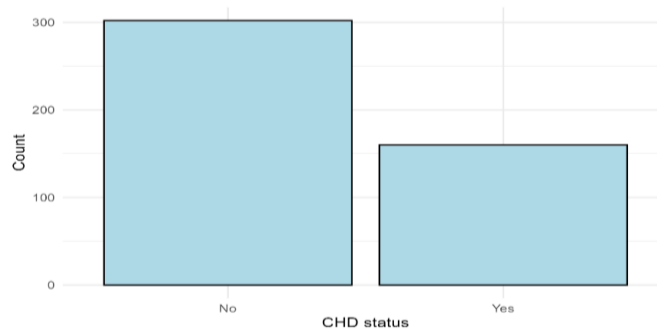


Figure 1: Class distribution of CHD outcome showing moderate imbalance.

Table 1: Distribution of CHD outcome

CHD status	Count	Percentage
No CHD (0)	302	65.4%
CHD (1)	160	34.6%

This imbalance is important because a naive model would achieve 65.4% accuracy, meaning additional metrics such as sensitivity and AUC are required.

2.3 Summary statistics

Several predictors, particularly tobacco and alcohol, exhibit right-skewed distributions with potential outliers, but exploratory checks suggested performance was adequate without transformation, so no transformations were applied.

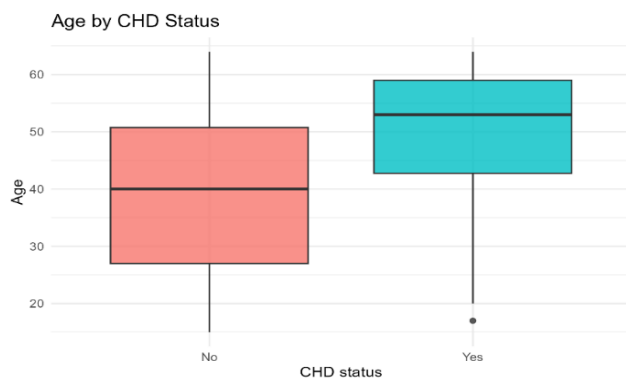
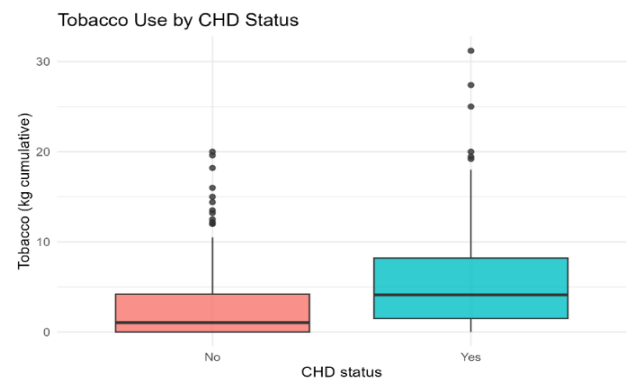
2.4 Comparison by disease status

A comparison of averages across CHD groups shows that patients with CHD are older and have less favourable cardiovascular risk profiles.

Table 2: Mean values by CHD status

Variable	No CHD	CHD
sbp	135.5	143.7
tobacco	2.63	5.52
ldl	4.34	5.49
adiposity	24.5	27.4
typea	52.3	54.8
obesity	25.6	26.8
alcohol	16.1	18.8
age	38.9	50.3

Patients with CHD are generally older and exhibit less favorable cardiovascular profiles, particularly in terms of tobacco use and LDL cholesterol. The largest difference is observed in age, with CHD patients being over 11 years older on average.

**Figure 2:** Age distribution by CHD status.**Figure 3:** Tobacco use by CHD status.

2.5 Family history

Family history is also strongly associated with CHD.

Table 3: Family history by CHD status

Family history	No CHD	CHD
Absent	206	64
Present	96	96

Among patients with CHD, **60.0%** have a positive family history, compared with only **31.8%** in the non-CHD group. This suggests it is an important predictor, although its binary nature may oversimplify underlying influences and may capture both genetic and shared environmental risk factors.

2.6 Correlation analysis

To examine linear relationships, a correlation matrix was computed using the numeric variables, with family history coded as binary. The correlation analysis identifies age as having the strongest linear association with CHD, followed by tobacco use, family history, and LDL cholesterol.

Table 4: Correlation with CHD

Variable	Correlation with CHD
age	0.373
tobacco	0.300
famhist	0.272

ldl	0.263
adiposity	0.254
sbp	0.193
typea	0.103
obesity	0.100
alcohol	0.066

Figure 4 shows that some predictors are correlated with each other, especially adiposity and obesity, and adiposity and age, suggesting multicollinearity may be present. This provides justification for the use of ridge regression, which stabilizes estimates by shrinking correlated coefficients.

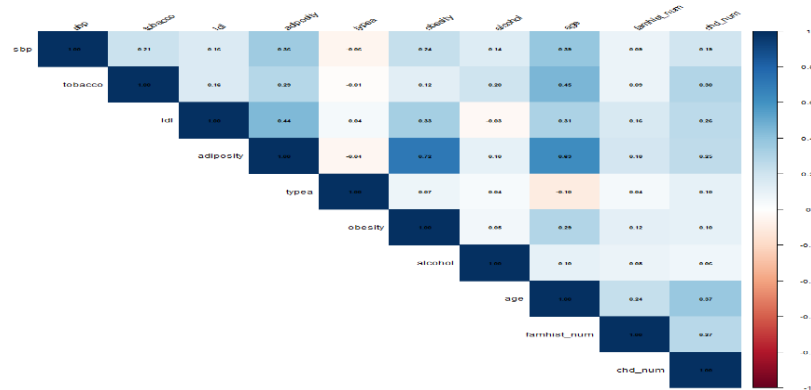


Figure 4: Correlation matrix showing relationships among predictors and CHD.

These patterns are consistent with established cardiovascular risk factors, suggesting that the predictors have meaningful clinical relevance and are likely to contribute to predictive performance.

3. Ridge Logistic Regression

3.1 Method

A **ridge logistic regression** model was fitted to classify CHD status. Since ridge regression penalises the size of coefficients, the predictors were standardised before fitting. The categorical variable **famhist** was converted into a binary indicator. The data were split into **70% training** and **30% test** sets using a stratified split so that the class proportions remained similar in both sets. The ridge penalty parameter λ was selected using grid search and **5-fold cross-validation** on the training data, with the optimal value chosen based on mean cross-validated accuracy. Ridge logistic regression minimizes a penalized log-likelihood that adds a $\lambda \sum \beta^2$ penalty to shrink coefficient magnitudes and reduce variance.

3.2 Model performance

The best ridge model achieved the following performance:

Table 5: Ridge logistic regression results

Measure	Value
Cross-validated training accuracy	0.727
Test accuracy	0.703
Test AUC	0.749

The relatively close agreement between cross-validated and test accuracy suggests that the model is not severely overfitting. The AUC of **0.749** indicates good discrimination ability between CHD and non-CHD

patients. The test accuracy of **0.703** exceeds the naive majority-class baseline of 0.654, confirming that the model improves on predicting the majority class alone.

3.3 Confusion matrix

Table 6: Confusion matrix for ridge logistic regression

	Predicted No CHD	Predicted CHD
Actual No CHD	75	15
Actual CHD	26	22

This indicates that the model achieves higher specificity than sensitivity. Sensitivity (true positive rate) is 0.458, while specificity (true negative rate) is 0.879. Adjusting the classification threshold or applying cost-sensitive methods could improve sensitivity. This imbalance between sensitivity and specificity highlights a limitation of the model, as failing to identify CHD cases (false negatives) may have serious clinical consequences.

3.4 Interpretation of ridge results

The ridge regression results indicate that age, tobacco use, LDL cholesterol, adiposity, and family history are positively associated with CHD risk, meaning higher values increase the probability of disease. Among the standardised predictors, age and cumulative tobacco exposure have the largest positive coefficients, implying the strongest marginal influence on the predicted CHD probability. In contrast, alcohol shows a relatively weak effect once other variables are accounted for. The correlation between obesity and adiposity suggests overlapping information, indicating the presence of multicollinearity. Unlike LASSO regression, ridge does not perform variable selection, meaning all predictors remain in the model regardless of their contribution, which may reduce interpretability.

4. Comparison with Other Classifiers

4.1 Procedure

To identify the highest-accuracy model, several alternative classifiers were tested using the same train-test split and 5-fold cross-validation framework. For each model, key hyperparameters (e.g. K for KNN, tree depth, number of trees, and SVM C and γ) were tuned using grid search with 5-fold cross-validation on the training data. The models considered were **K-nearest neighbours (KNN)**, **linear discriminant analysis (LDA)**, **decision tree**, **random forest**, and **support vector machine with radial kernel (SVM)**, and all models used the same predictors. Methods that are sensitive to variable scale were standardised before fitting.

4.2 Results

Table 7: Comparison of classifiers

Model	Test Accuracy	Test AUC
Ridge Logistic Regression	0.703	0.749
KNN	0.725	0.710
LDA	0.703	0.750
Decision Tree	0.696	0.605
Random Forest	0.717	0.711
SVM (RBF)	0.717	0.726

The decision tree performs worst, and random forest gives only modest gains, suggesting limited benefit from more complex tree-based models on this dataset. Given the class imbalance, AUC provides a more reliable basis for comparing models than accuracy alone, and it was therefore prioritized when selecting the final classifier.

4.3 Best classifier

KNN achieved the highest test accuracy (0.725), with random forest and SVM performing similarly. However, ridge logistic regression and LDA produced the highest AUC values (0.749 and 0.750), indicating better discrimination between CHD and non-CHD cases. Although KNN performs best in terms of accuracy, ridge logistic regression is preferred as the final model due to its stronger discrimination, robustness to multicollinearity, and interpretability. These features make it more suitable for understanding risk factors and for use in a clinical context.

5. Discussion

A key feature of this dataset is the presence of correlated predictors, particularly adiposity and obesity, which introduces multicollinearity. This can lead to unstable coefficient estimates in standard logistic regression, making ridge regression appropriate as it shrinks coefficients and improves model stability. Class imbalance is also evident, with only around one-third of patients having CHD, contributing to higher specificity than sensitivity. This is a limitation in a clinical context, where failing to detect CHD (false negatives) may have serious consequences. Adjusting the classification threshold or applying resampling or cost-sensitive methods could improve sensitivity. The absence of interaction terms may limit the ability to capture combined risk effects, such as the joint influence of age and smoking. However, the similar performance of linear and more flexible models suggests that relationships are largely linear, reducing the benefit of more complex methods. Finally, the relatively small sample size ($n = 462$), restriction to male patients from a single region, and reliance on a single train–test split limit generalisability and introduce uncertainty in performance estimates. The limited improvement from more complex models suggests that predictive performance is constrained more by data size and quality than by model choice. Future work could explore interaction effects and alternative classification threshold to improve clinical applicability.

6. Conclusion

This report analysed a heart-disease dataset using exploratory data analysis, ridge logistic regression, and a range of classification methods. While KNN achieved the highest accuracy, ridge logistic regression and LDA demonstrated superior discrimination based on AUC. Key predictors included age, tobacco use, LDL cholesterol, and family history. However, the relatively low sensitivity highlights a limitation for clinical application, indicating that further refinement is required.

Appendix

```
# HEART DISEASE CLASSIFICATION COURSEWORK

# 0. Clear workspace
# -----
rm(list = ls())

# -----
# 1. Install packages if needed
# -----
# Uncomment if you need to install them:
# install.packages(c(
#   "tidyverse", "caret", "glmnet", "MASS", "rpart",
#   "rpart.plot", "randomForest", "e1071", "pROC", "corrplot"
# ))

# -----
# 2. Load packages
# -----
library(MASS)
library(tidyverse)

## — Attaching core tidyverse packages ————— tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    4.0.2      ✓ tibble     3.2.1
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.0.4
## — Conflicts ————— tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()    masks stats::lag()
## ✗ dplyr::select() masks MASS::select()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
## become errors

library(caret)

## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift

library(glmnet)

## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Loaded glmnet 4.1-8
```

```

library(rpart)
library(rpart.plot)
library(randomForest)

## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:dplyr':
##
##      combine
##
## The following object is masked from 'package:ggplot2':
##
##      margin

library(e1071)

##
## Attaching package: 'e1071'
##
## The following object is masked from 'package:ggplot2':
##
##      element

library(pROC)

## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
##
## The following objects are masked from 'package:stats':
##
##      cov, smooth, var

library(corrplot)

## corrplot 0.95 loaded

# -----
# 3. Read dataset
# -----
# Make sure heart-disease.csv is in your working directory
heart <- read.csv("heart-disease.csv", stringsAsFactors = FALSE)

# -----
# 4. Prepare data
# -----
heart$famhist <- factor(heart$famhist, levels = c("Absent", "Present"))
heart$chd <- factor(heart$chd, levels = c(0, 1), labels = c("No", "Yes"))

# -----
# 5. Initial inspection
# -----
cat("Dimensions of dataset:\n")

## Dimensions of dataset:
print(dim(heart))

```

```
## [1] 462 10

cat("\nVariable names:\n")

##
## Variable names:

print(names(heart))

## [1] "sbp"      "tobacco"   "ldl"       "adiposity" "famhist"   "typea"
## [7] "obesity"  "alcohol"   "age"       "chd"

cat("\nStructure of dataset:\n")

##
## Structure of dataset:

str(heart)

## 'data.frame': 462 obs. of 10 variables:
## $ sbp : int 160 144 118 170 134 132 142 114 114 132 ...
## $ tobacco : num 12 0.01 0.08 7.5 13.6 6.2 4.05 4.08 0 0 ...
## $ ldl : num 5.73 4.41 3.48 6.41 3.5 6.47 3.38 4.59 3.83 5.8 ...
## $ adiposity: num 23.1 28.6 32.3 38 27.8 ...
## $ famhist : Factor w/ 2 levels "Absent","Present": 2 1 2 2 2 2 1 2 2 2 ...
## $ typea : int 49 55 52 51 60 62 59 62 49 69 ...
## $ obesity : num 25.3 28.9 29.1 32 26 ...
## $ alcohol : num 97.2 2.06 3.81 24.26 57.34 ...
## $ age : int 52 63 46 58 49 45 38 58 29 53 ...
## $ chd : Factor w/ 2 levels "No","Yes": 2 2 1 2 2 1 1 2 1 2 ...

cat("\nSummary statistics:\n")

##
## Summary statistics:

print(summary(heart))

## sbp tobacco ldl adiposity
## Min. :101.0 Min. : 0.0000 Min. : 0.980 Min. : 6.74
## 1st Qu.:124.0 1st Qu.: 0.0525 1st Qu.: 3.283 1st Qu.:19.77
## Median :134.0 Median : 2.0000 Median : 4.340 Median :26.11
## Mean :138.3 Mean : 3.6356 Mean : 4.740 Mean :25.41
## 3rd Qu.:148.0 3rd Qu.: 5.5000 3rd Qu.: 5.790 3rd Qu.:31.23
## Max. :218.0 Max. :31.2000 Max. :15.330 Max. :42.49
## famhist typea obesity alcohol age
## Absent:270 Min. :13.0 Min. :14.70 Min. : 0.00 Min. :15.00
## Present:192 1st Qu.:47.0 1st Qu.:22.98 1st Qu.: 0.51 1st Qu.:31.00
## Median :53.0 Median :25.80 Median : 7.51 Median :45.00
## Mean :53.1 Mean :26.04 Mean : 17.04 Mean :42.82
## 3rd Qu.:60.0 3rd Qu.:28.50 3rd Qu.: 23.89 3rd Qu.:55.00
## Max. :78.0 Max. :46.58 Max. :147.19 Max. :64.00
## chd
## No :302
## Yes:160
##
##
cat("\nMissing values:\n")
```

```
##
## Missing values:

print(colSums(is.na(heart)))

##          sbp    tobacco      ldl adiposity    famhist      typea    obesity    alcohol
##          0         0         0         0         0         0         0         0
##        age      chd
##          0         0

# -----
# 6. Create output folder
# -----
if (!dir.exists("figures")) {
  dir.create("figures")
}

# -----
# 7. Exploratory Data Analysis
# -----

# 7.1 CHD class balance
cat("\nCHD frequency table:\n")

##
## CHD frequency table:

print(table(heart$chd))

##
##   No Yes
## 302 160

cat("\nCHD proportion table:\n")

##
## CHD proportion table:

print(prop.table(table(heart$chd)))

##
##           No           Yes
## 0.6536797 0.3463203

# 7.2 Summary by CHD group
group_summary <- heart %>%
  dplyr::group_by(chd) %>%
  dplyr::summarise(
    n = dplyr::n(),
    mean_sbp = mean(sbp),
    mean_tobacco = mean(tobacco),
    mean_ldl = mean(ldl),
    mean_adiposity = mean(adiposity),
    mean_typea = mean(typea),
    mean_obesity = mean(obesity),
    mean_alcohol = mean(alcohol),
    mean_age = mean(age),
    .groups = "drop"
  )

cat("\nSummary by CHD group:\n")
```

```
##
## Summary by CHD group:

print(group_summary)

## # A tibble: 2 × 10
##   chd      n mean_sbp mean_tobacco mean_ldl mean_adiposity mean_typea
##   <fct> <int>   <dbl>       <dbl>   <dbl>       <dbl>       <dbl>
## 1 No     302    135.        2.63     4.34        24.0        52.4
## 2 Yes    160    144.        5.52     5.49        28.1        54.5
## # i 3 more variables: mean_obesity <dbl>, mean_alcohol <dbl>, mean_age <dbl>

# 7.3 Family history by CHD
fh_table <- table(heart$famhist, heart$chd)

cat("\nFamily history by CHD:\n")

##
## Family history by CHD:

print(fh_table)

##
##           No Yes
## Absent   206  64
## Present   96  96

cat("\nColumn proportions for family history by CHD:\n")

##
## Column proportions for family history by CHD:

print(prop.table(fh_table, margin = 2))

##
##           No      Yes
## Absent  0.6821192 0.4000000
## Present 0.3178808 0.6000000

# 7.4 Overall summary statistics
overall_summary <- heart %>%
  dplyr::summarise(
    mean_sbp = mean(sbp), median_sbp = median(sbp), sd_sbp = sd(sbp),
    mean_tobacco = mean(tobacco), median_tobacco = median(tobacco), sd_tobacco = sd(tobacco),
    mean_ldl = mean(ldl), median_ldl = median(ldl), sd_ldl = sd(ldl),
    mean_adiposity = mean(adiposity), median_adiposity = median(adiposity), sd_adiposity = sd(adiposity),
    mean_typea = mean(typea), median_typea = median(typea), sd_typea = sd(typea),
    mean_obesity = mean(obesity), median_obesity = median(obesity), sd_obesity = sd(obesity),
    mean_alcohol = mean(alcohol), median_alcohol = median(alcohol), sd_alcohol = sd(alcohol),
    mean_age = mean(age), median_age = median(age), sd_age = sd(age)
  )

cat("\nOverall summary statistics:\n")

##
## Overall summary statistics:

print(overall_summary)
```

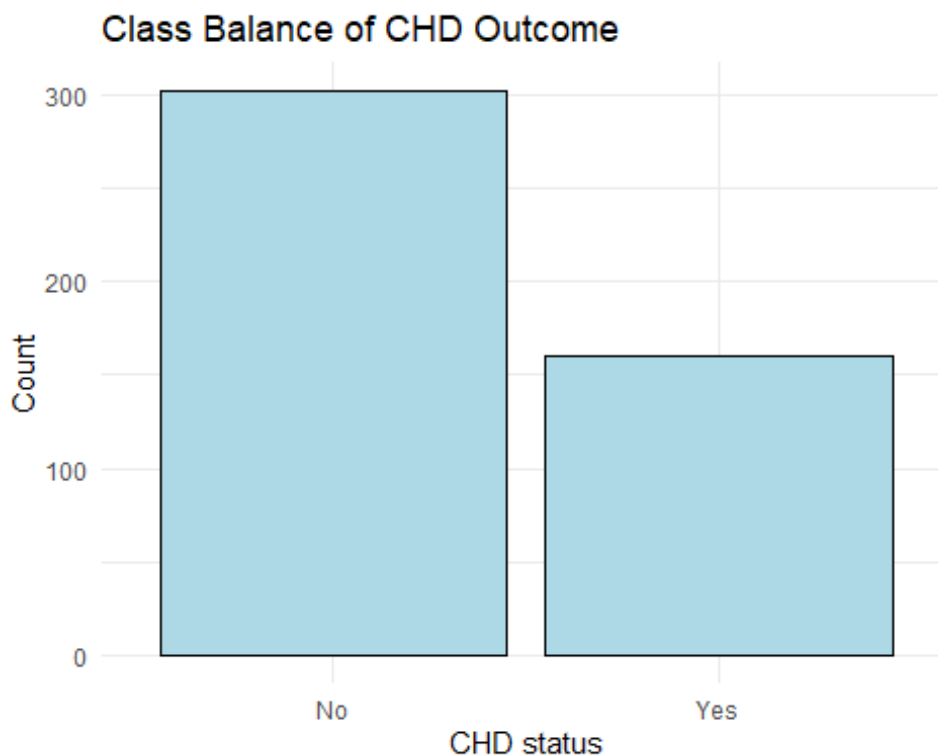
```
## mean_sbp median_sbp sd_sbp mean_tobacco median_tobacco sd_tobacco mean_ldl
## 1 138.3268 134 20.49632 3.635649 2 4.593024 4.740325
## median_ldl sd_ldl mean_adiposity median_adiposity sd_adiposity mean_typea
## 1 4.34 2.070909 25.40673 26.115 7.780699 53.1039
## median_typea sd_typea mean_obesity median_obesity sd_obesity mean_alcohol
## 1 53 9.817534 26.04411 25.805 4.21368 17.04439
## median_alcohol sd_alcohol mean_age median_age sd_age
## 1 7.51 24.48106 42.81602 45 14.60896
```

```
# -----
# 8. Figures for EDA
# -----
```

```
# Figure 1: Class balance
```

```
p1 <- ggplot(heart, aes(x = chd)) +
  geom_bar(fill = "lightblue", color = "black") +
  labs(
    title = "Class Balance of CHD Outcome",
    x = "CHD status",
    y = "Count"
  ) +
  theme_minimal()
```

```
print(p1)
```

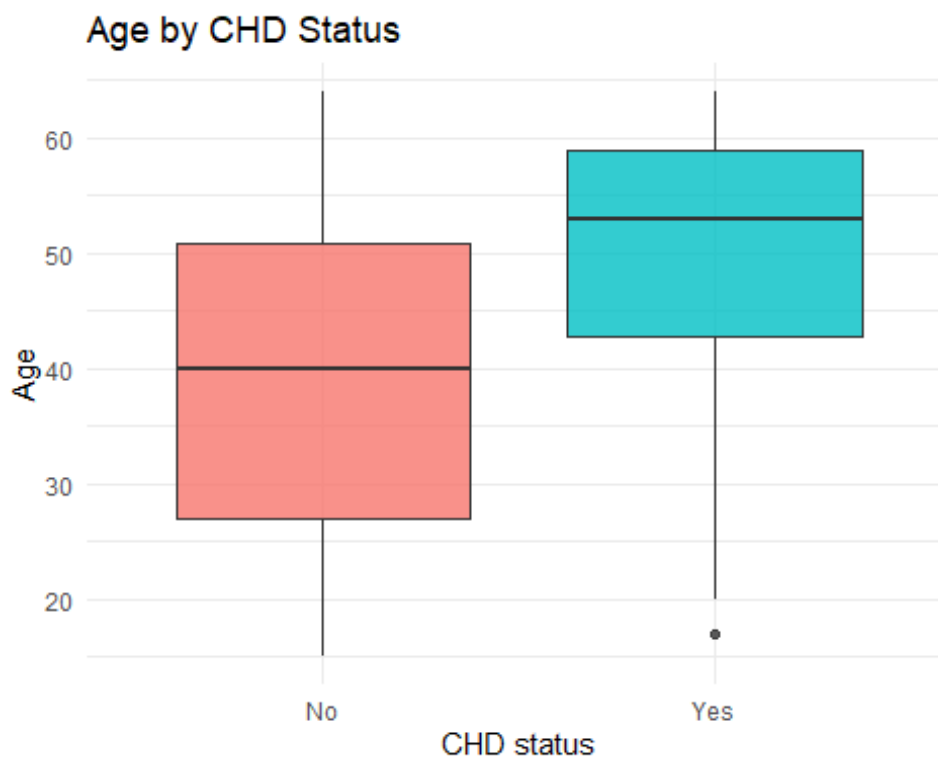


```
ggsave("figures/figure1_class_balance.png", plot = p1, width = 6, height = 4, dpi = 300)
```

```
# Figure 2: Age by CHD
```

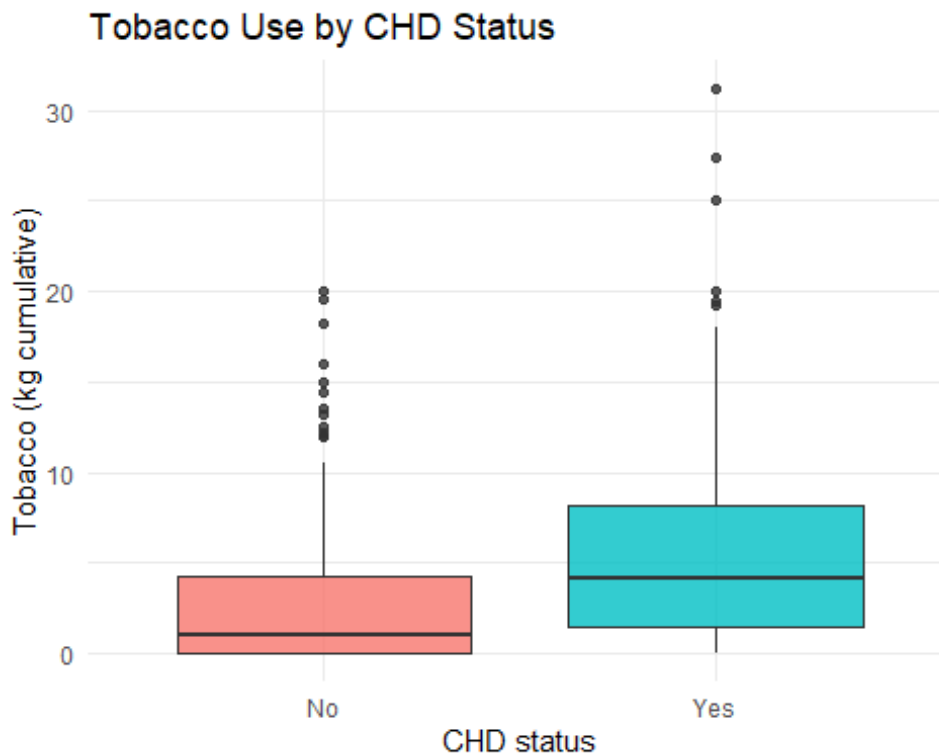
```
p2 <- ggplot(heart, aes(x = chd, y = age, fill = chd)) +
  geom_boxplot(alpha = 0.8) +
  labs(
    title = "Age by CHD Status",
    x = "CHD status",
    y = "Age"
  ) +
  theme_minimal() +
```

```
theme(legend.position = "none")
print(p2)
```



```
ggsave("figures/figure2_age_boxplot.png", plot = p2, width = 6, height = 4, dpi = 300)
```

```
# Figure 3: Tobacco by CHD
p3 <- ggplot(heart, aes(x = chd, y = tobacco, fill = chd)) +
  geom_boxplot(alpha = 0.8) +
  labs(
    title = "Tobacco Use by CHD Status",
    x = "CHD status",
    y = "Tobacco (kg cumulative)"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
print(p3)
```



```
ggsave("figures/figure3_tobacco_boxplot.png", plot = p3, width = 6, height = 4, dpi = 300)
```

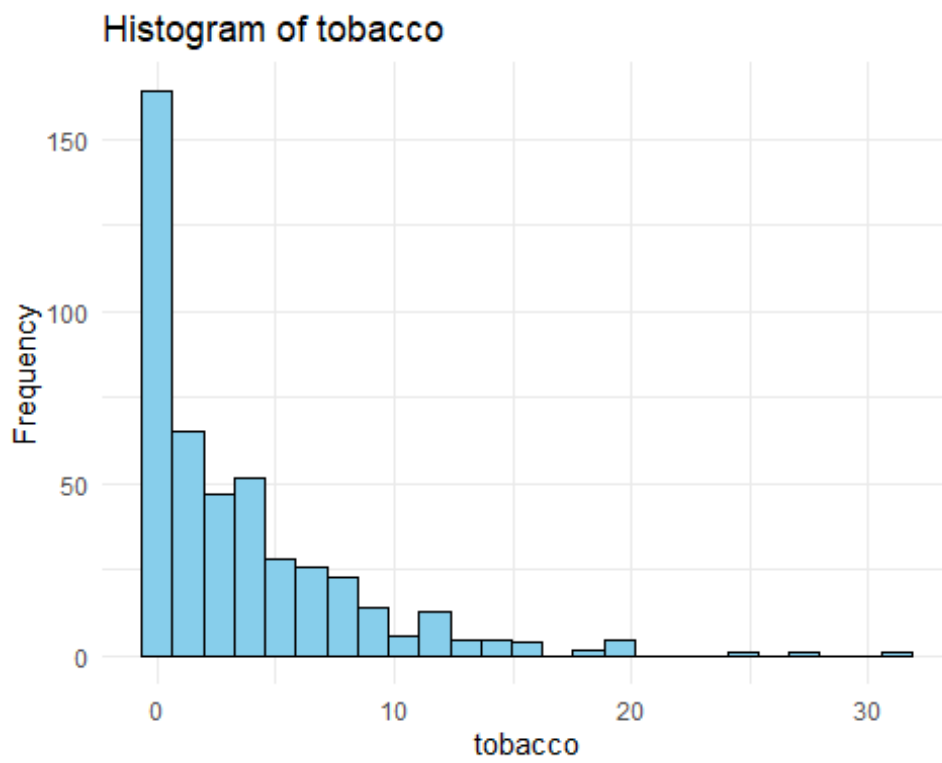
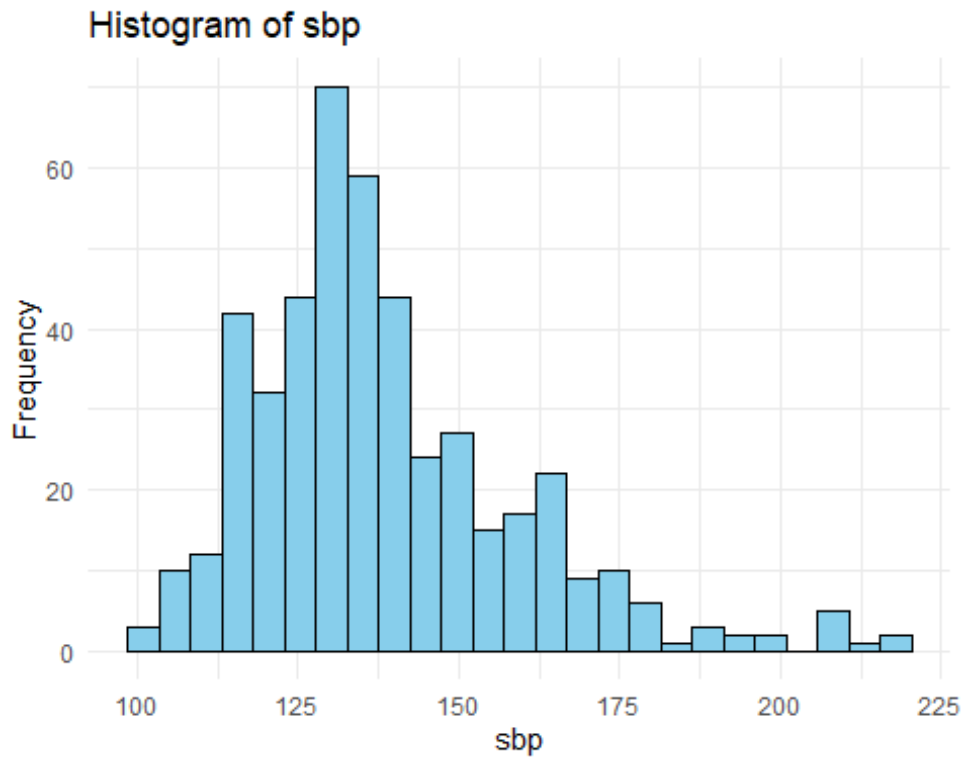
Optional: histograms for all continuous variables

```
num_vars <- c("sbp", "tobacco", "ldl", "adiposity", "typea", "obesity", "alcohol", "age")
```

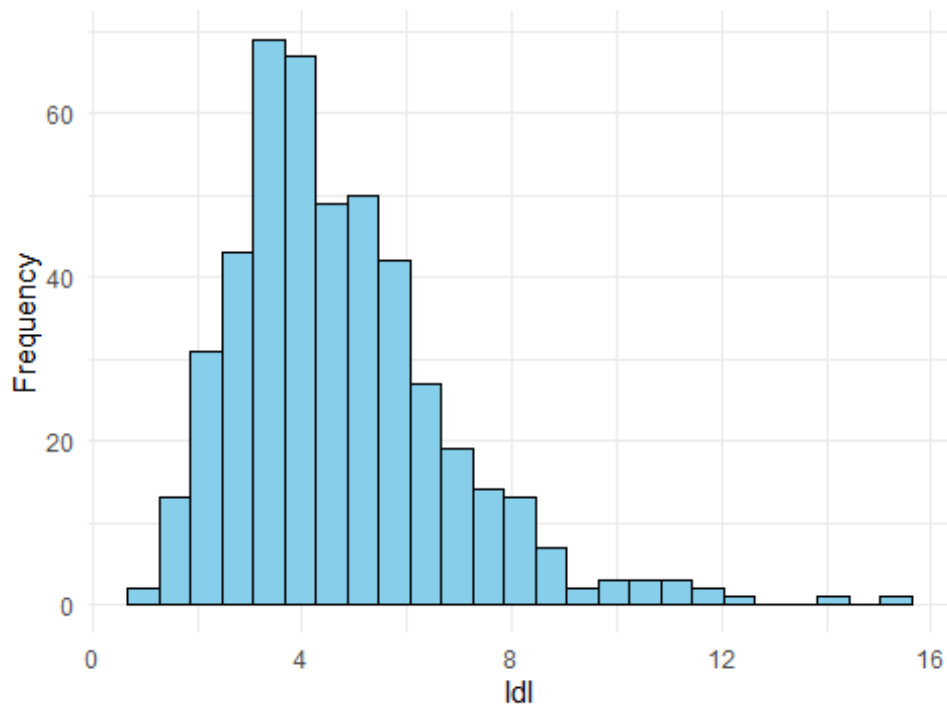
```
for (v in num_vars) {
  p <- ggplot(heart, aes_string(x = v)) +
    geom_histogram(bins = 25, fill = "skyblue", color = "black") +
    labs(
      title = paste("Histogram of", v),
      x = v,
      y = "Frequency"
    ) +
    theme_minimal()
}
```

```
print(p)
ggsave(
  filename = paste0("figures/hist_", v, ".png"),
  plot = p,
  width = 6,
  height = 4,
  dpi = 300
)
```

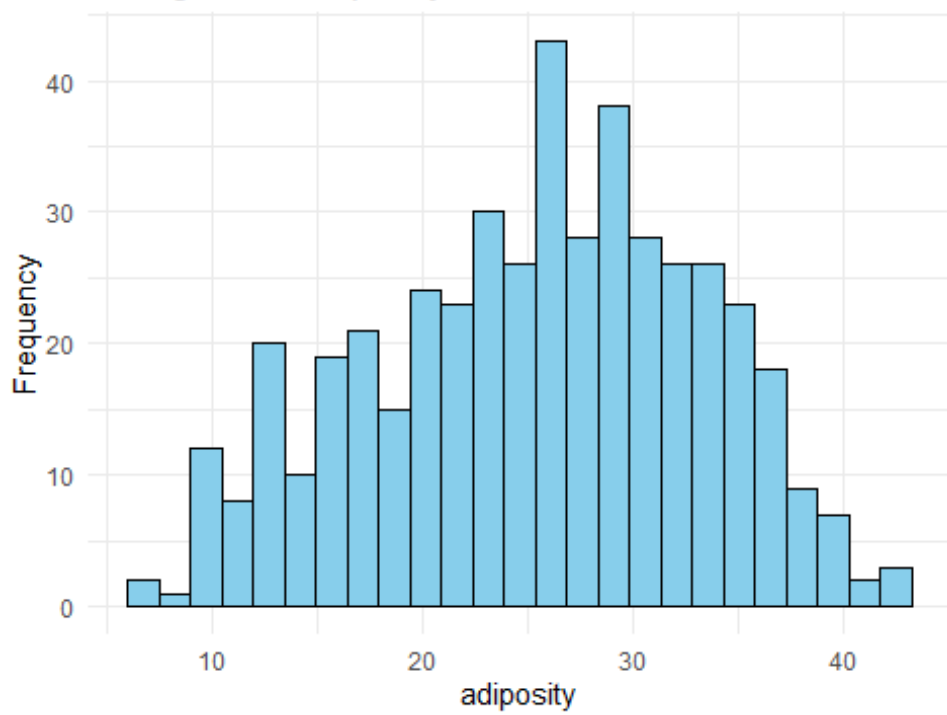
```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with `aes()`.
## i See also `vignette("ggplot2-in-packages")` for more information.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

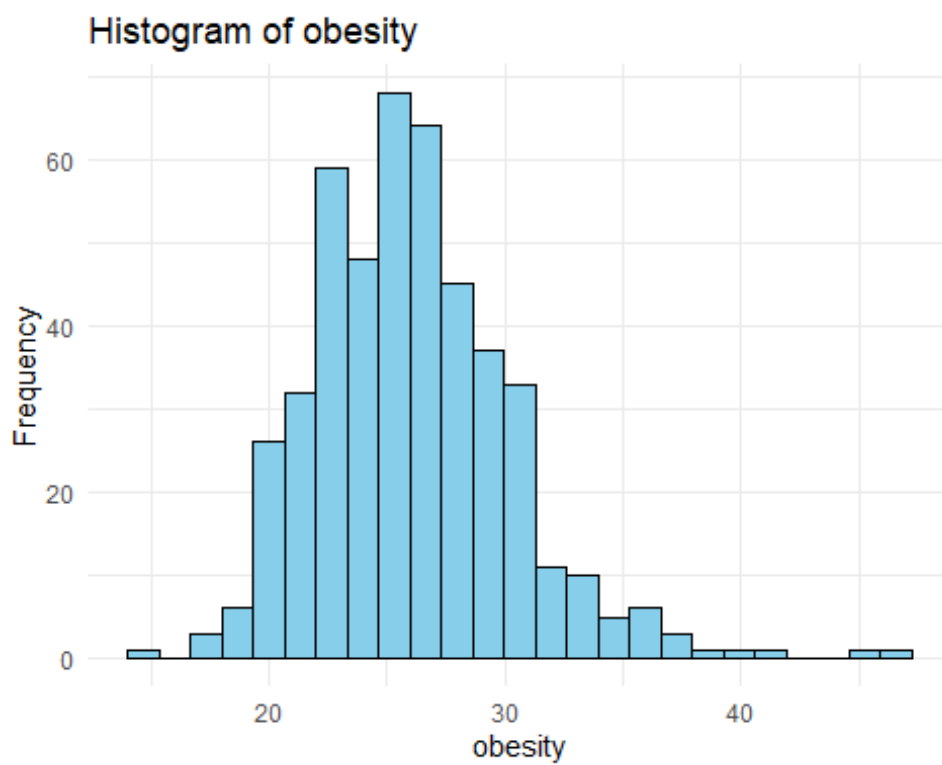
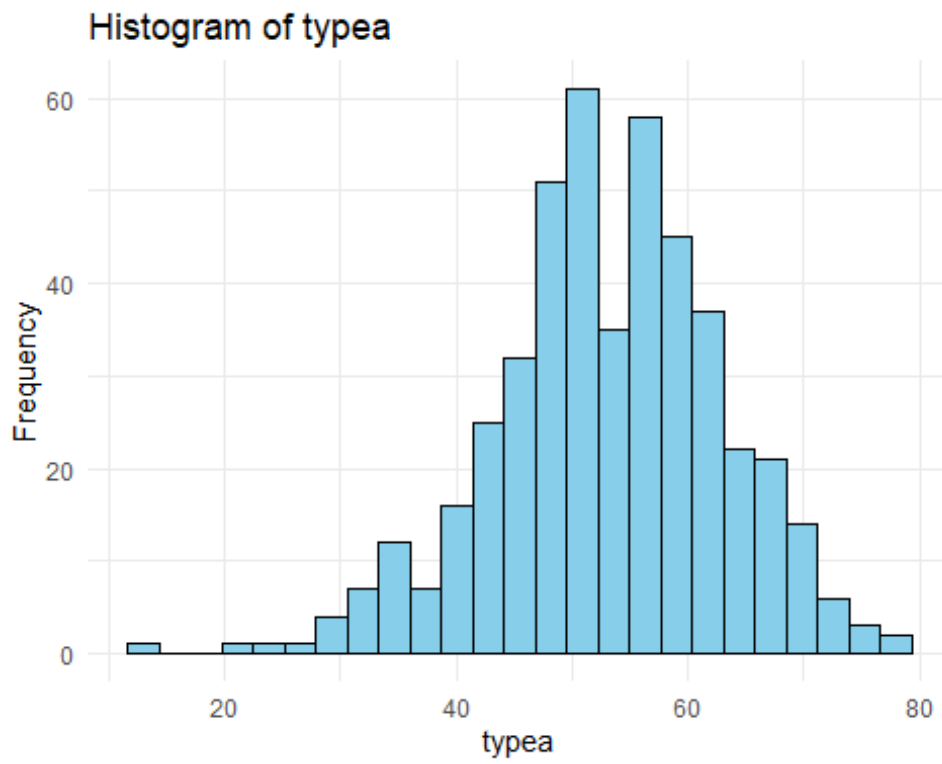


Histogram of ldl

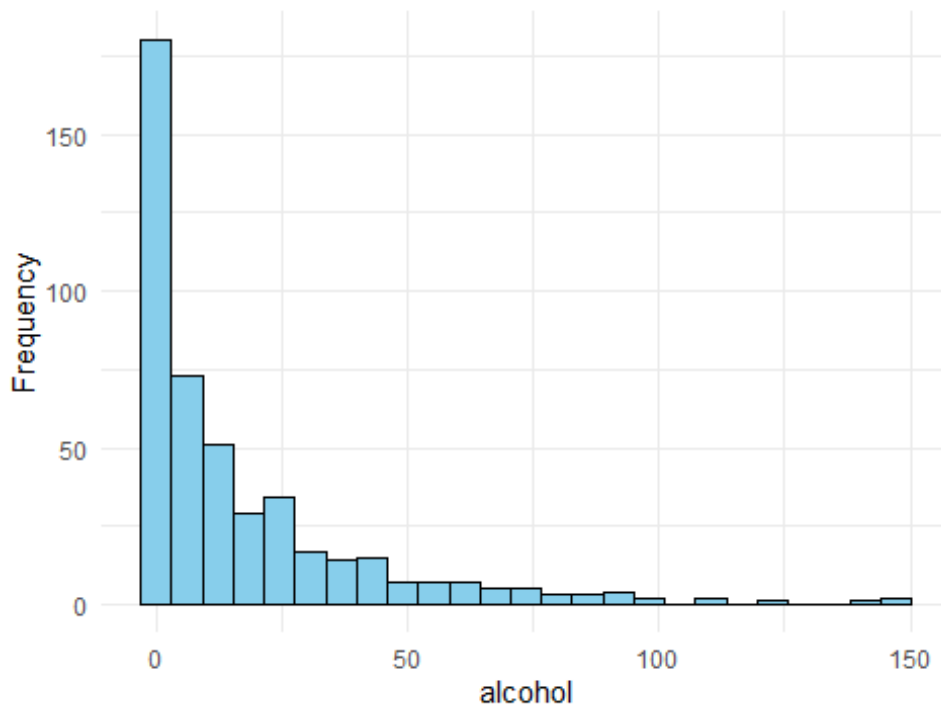


Histogram of adiposity

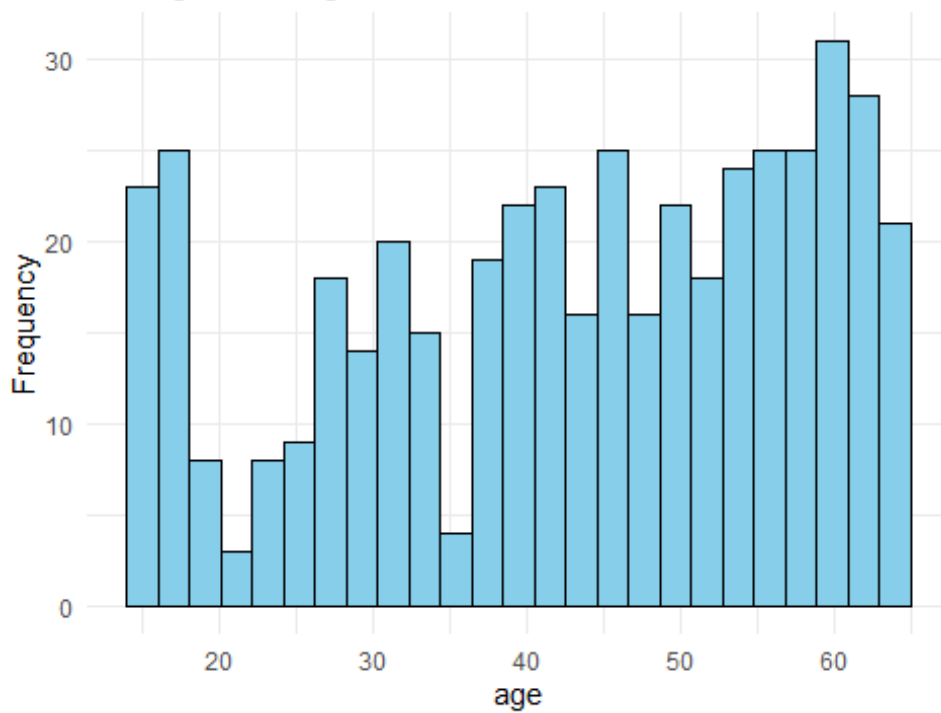




Histogram of alcohol



Histogram of age



```
# -----
# 9. Figure 4: Correlation Heatmap
# -----
# Convert famhist and chd into numeric variables for the correlation matrix
heart_num <- heart %>%
  dplyr::mutate(
    famhist_num = ifelse(famhist == "Present", 1, 0),
    chd_num = ifelse(chd == "Yes", 1, 0)
  ) %>%
  dplyr::select(
    sbp, tobacco, ldl, adiposity, typea,
    obesity, alcohol, age, famhist_num, chd_num
  )
```

```

# Correlation matrix
cor_matrix <- cor(heart_num)

cat("\nCorrelation matrix:\n")

##
## Correlation matrix:

print(round(cor_matrix, 3))

##           sbp tobacco   ldl adiposity  typea obesity alcohol   age
## sbp       1.000   0.212  0.158   0.357 -0.057   0.238   0.140  0.389
## tobacco   0.212   1.000  0.159   0.287 -0.015   0.125   0.201  0.450
## ldl       0.158   0.159  1.000   0.440  0.044   0.331  -0.033  0.312
## adiposity 0.357   0.287  0.440   1.000 -0.043   0.717   0.100  0.626
## typea     -0.057 -0.015  0.044  -0.043  1.000   0.074   0.039 -0.103
## obesity   0.238   0.125  0.331   0.717  0.074   1.000   0.052  0.292
## alcohol   0.140   0.201 -0.033   0.100  0.039   0.052   1.000  0.101
## age       0.389   0.450  0.312   0.626 -0.103   0.292   0.101  1.000
## famhist_num 0.086  0.089  0.161   0.182  0.045   0.116   0.081  0.240
## chd_num    0.192  0.300  0.263   0.254  0.103   0.100   0.063  0.373
##           famhist_num chd_num
## sbp           0.086   0.192
## tobacco       0.089   0.300
## ldl           0.161   0.263
## adiposity     0.182   0.254
## typea         0.045   0.103
## obesity       0.116   0.100
## alcohol       0.081   0.063
## age          0.240   0.373
## famhist_num   1.000   0.272
## chd_num       0.272   1.000

# Save numeric correlation matrix
write.csv(as.data.frame.matrix(round(cor_matrix, 3)), "correlation_matrix.csv")

# Save Figure 4 using corrplot
png("figures/figure4_correlation_heatmap.png", width = 1000, height = 900)
corrplot(
  cor_matrix,
  method = "color",
  type = "upper",
  addCoef.col = "black",
  number.cex = 0.7,
  tl.col = "black",
  tl.srt = 45,
  diag = TRUE,
  mar = c(0, 0, 2, 0),
  title = "Correlation Heatmap"
)
dev.off()

## png
## 2

# Also print it in the plotting window
corrplot(
  cor_matrix,
  method = "color",

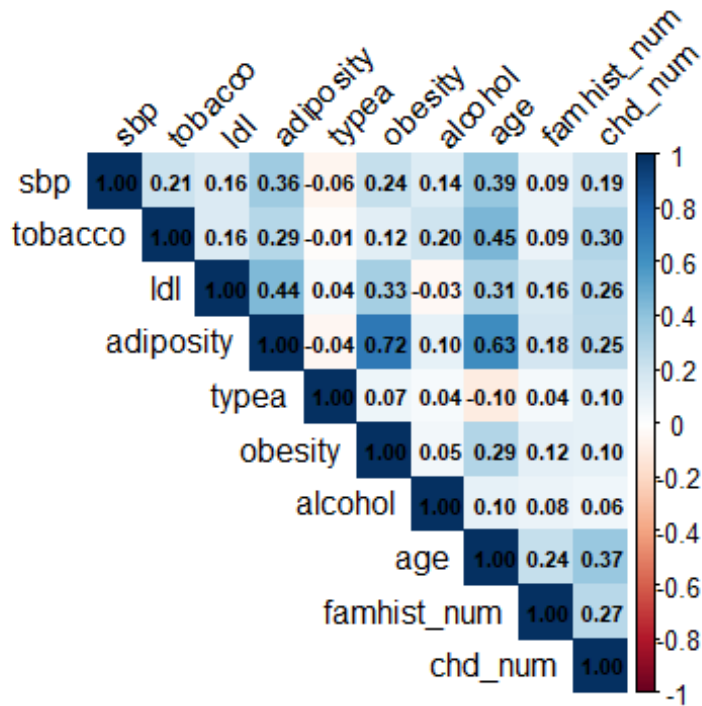
```

```

type = "upper",
addCoef.col = "black",
number.cex = 0.7,
tl.col = "black",
tl.srt = 45,
diag = TRUE,
mar = c(0, 0, 2, 0),
title = "Figure 4. Correlation Heatmap"
)

```

Figure 4. Correlation Heatmap



```

# -----
# 10. Train-test split
# -----
set.seed(123)

train_index <- createDataPartition(heart$chd, p = 0.7, list = FALSE)
train_data <- heart[train_index, ]
test_data <- heart[-train_index, ]

cat("\nTraining set size:\n")

##
## Training set size:
print(dim(train_data))

## [1] 324 10

cat("\nTest set size:\n")

##
## Test set size:
print(dim(test_data))

## [1] 138 10

cat("\nTraining CHD proportions:\n")

```

```
##
## Training CHD proportions:

print(prop.table(table(train_data$chd)))

##
##           No           Yes
## 0.654321 0.345679

cat("\nTest CHD proportions:\n")

##
## Test CHD proportions:

print(prop.table(table(test_data$chd)))

##
##           No           Yes
## 0.6521739 0.3478261

# -----
# 11. Ridge Logistic Regression
# -----
x_train <- model.matrix(chd ~ ., data = train_data)[, -1]
y_train <- ifelse(train_data$chd == "Yes", 1, 0)

x_test <- model.matrix(chd ~ ., data = test_data)[, -1]
y_test <- ifelse(test_data$chd == "Yes", 1, 0)

set.seed(123)
cv_ridge <- cv.glmnet(
  x = x_train,
  y = y_train,
  family = "binomial",
  alpha = 0,
  nfolds = 5,
  standardize = TRUE,
  type.measure = "class"
)

cat("\nBest lambda for ridge (lambda.min):\n")

##
## Best lambda for ridge (lambda.min):

print(cv_ridge$lambda.min)

## [1] 0.07300937

cat("\n1-SE lambda for ridge (lambda.1se):\n")

##
## 1-SE lambda for ridge (lambda.1se):

print(cv_ridge$lambda.1se)

## [1] 0.1400254

png("figures/ridge_cv_plot.png", width = 800, height = 600)
plot(cv_ridge)
title("Cross-Validation Plot for Ridge Logistic Regression")
dev.off()
```

```

## png
## 2

ridge_model <- glmnet(
  x = x_train,
  y = y_train,
  family = "binomial",
  alpha = 0,
  lambda = cv_ridge$lambda.min,
  standardize = TRUE
)

cat("\nRidge coefficients:\n")

##
## Ridge coefficients:

print(coef(ridge_model))

## 10 x 1 sparse Matrix of class "dgCMatrix"
##              s0
## (Intercept) -4.979758e+00
## sbp          2.856332e-03
## tobacco     7.202359e-02
## ldl          1.247560e-01
## adiposity    3.027757e-02
## famhistPresent 6.846772e-01
## typea        2.653436e-02
## obesity      -3.053389e-02
## alcohol      -4.840931e-05
## age          2.919533e-02

ridge_prob <- predict(ridge_model, newx = x_test, type = "response")[, 1]
ridge_pred <- ifelse(ridge_prob > 0.5, "Yes", "No")
ridge_pred <- factor(ridge_pred, levels = c("No", "Yes"))

cat("\nConfusion matrix for ridge logistic regression:\n")

##
## Confusion matrix for ridge logistic regression:

ridge_cm <- confusionMatrix(ridge_pred, test_data$chd, positive = "Yes")
print(ridge_cm)

## Confusion Matrix and Statistics
##
##              Reference
## Prediction No Yes
##      No  75  26
##      Yes 15  22
##
##              Accuracy : 0.7029
##              95% CI : (0.6192, 0.7776)
##      No Information Rate : 0.6522
##      P-Value [Acc > NIR] : 0.1219
##
##              Kappa : 0.3081
##
##      McNemar's Test P-Value : 0.1183
##
##              Sensitivity : 0.4583

```

```

##           Specificity : 0.8333
##           Pos Pred Value : 0.5946
##           Neg Pred Value : 0.7426
##           Prevalence : 0.3478
##           Detection Rate : 0.1594
##           Detection Prevalence : 0.2681
##           Balanced Accuracy : 0.6458
##
##           'Positive' Class : Yes
##

ridge_acc <- as.numeric(ridge_cm$overall["Accuracy"])
cat("\nRidge test accuracy:\n")

##
## Ridge test accuracy:

print(ridge_acc)

## [1] 0.7028986

ridge_roc <- roc(response = test_data$chd, predictor = ridge_prob, levels = c("No", "Yes")
)

## Setting direction: controls < cases

ridge_auc <- as.numeric(auc(ridge_roc))

cat("\nRidge test AUC:\n")

##
## Ridge test AUC:

print(ridge_auc)

## [1] 0.7486111

png("figures/ridge_roc_curve.png", width = 800, height = 600)
plot(ridge_roc, main = "ROC Curve - Ridge Logistic Regression")
dev.off()

## png
## 2

# -----
# 12. Other classifiers
# -----

ctrl <- trainControl(
  method = "cv",
  number = 5,
  classProbs = TRUE,
  summaryFunction = twoClassSummary,
  savePredictions = "final"
)

# KNN
set.seed(123)
knn_model <- train(
  chd ~ .,
  data = train_data,
  method = "knn",
  preProcess = c("center", "scale"),

```

```

tuneGrid = expand.grid(k = c(3, 5, 7, 9, 11, 15, 21)),
trControl = ctrl,
metric = "ROC"
)

cat("\nKNN model:\n")

##
## KNN model:

print(knn_model)

## k-Nearest Neighbors
##
## 324 samples
## 9 predictor
## 2 classes: 'No', 'Yes'
##
## Pre-processing: centered (9), scaled (9)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 260, 259, 259, 260, 258
## Resampling results across tuning parameters:
##
##  k    ROC          Sens          Spec
##  3  0.6302125  0.7877076  0.3577075
##  5  0.6795763  0.8022148  0.3830040
##  7  0.7050053  0.8303433  0.3924901
##  9  0.7220217  0.8633444  0.4197628
## 11  0.7392228  0.8584718  0.4379447
## 15  0.7468399  0.8867110  0.4098814
## 21  0.7743223  0.9054264  0.3924901
##
## ROC was used to select the optimal model using the largest value.
## The final value used for the model was k = 21.

# LDA
set.seed(123)
lda_model <- train(
  chd ~ .,
  data = train_data,
  method = "lda",
  preProcess = c("center", "scale"),
  trControl = ctrl,
  metric = "ROC"
)

cat("\nLDA model:\n")

##
## LDA model:

print(lda_model)

## Linear Discriminant Analysis
##
## 324 samples
## 9 predictor
## 2 classes: 'No', 'Yes'
##
## Pre-processing: centered (9), scaled (9)

```

```
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 260, 259, 259, 260, 258
## Resampling results:
##
##      ROC          Sens          Spec
##  0.7903363  0.8723145  0.5173913

# Decision Tree
set.seed(123)
tree_model <- train(
  chd ~ .,
  data = train_data,
  method = "rpart",
  tuneLength = 10,
  trControl = ctrl,
  metric = "ROC"
)

cat("\nDecision Tree model:\n")

##
## Decision Tree model:

print(tree_model)

## CART
##
## 324 samples
##   9 predictor
##   2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 260, 259, 259, 260, 258
## Resampling results across tuning parameters:
##
##      cp          ROC          Sens          Spec
##  0.00000000  0.6961747  0.7925803  0.3920949
##  0.01587302  0.6967900  0.8021041  0.4185771
##  0.03174603  0.6423452  0.8255814  0.3747036
##  0.04761905  0.6576380  0.8490587  0.4391304
##  0.06349206  0.6576380  0.8490587  0.4391304
##  0.07936508  0.5858189  0.8871539  0.2798419
##  0.09523810  0.5845044  0.8638981  0.3146245
##  0.11111111  0.5845044  0.8638981  0.3146245
##  0.12698413  0.5822846  0.8499446  0.3328063
##  0.14285714  0.5584750  0.9023256  0.2146245
##
## ROC was used to select the optimal model using the largest value.
## The final value used for the model was cp = 0.01587302.

png("figures/decision_tree.png", width = 900, height = 700)
rpart.plot(tree_model$finalModel, main = "Decision Tree")
dev.off()

## png
##   2

# Random Forest
set.seed(123)
```

```

rf_model <- train(
  chd ~ .,
  data = train_data,
  method = "rf",
  tuneGrid = expand.grid(mtry = c(2, 3, 4, 5)),
  ntree = 300,
  trControl = ctrl,
  metric = "ROC"
)

cat("\nRandom Forest model:\n")

##
## Random Forest model:

print(rf_model)

## Random Forest
##
## 324 samples
## 9 predictor
## 2 classes: 'No', 'Yes'
##
## No pre-processing
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 260, 259, 259, 260, 258
## Resampling results across tuning parameters:
##
## mtry ROC Sens Spec
## 2 0.7459535 0.8399779 0.4193676
## 3 0.7353463 0.8352159 0.4106719
## 4 0.7296460 0.8211517 0.4363636
## 5 0.7282072 0.8214839 0.4189723
##
## ROC was used to select the optimal model using the largest value.
## The final value used for the model was mtry = 2.

rf_varimp <- varImp(rf_model)
cat("\nRandom Forest variable importance:\n")

##
## Random Forest variable importance:

print(rf_varimp)

## rf variable importance
##
## Overall
## age 100.00
## tobacco 91.40
## ldl 78.30
## adiposity 69.57
## typea 57.37
## obesity 53.99
## sbp 50.96
## alcohol 33.99
## famhistPresent 0.00

png("figures/random_forest_varimp.png", width = 800, height = 600)
plot(rf_varimp, main = "Variable Importance - Random Forest")
dev.off()

```

```
## png
## 2

# SVM with radial kernel
set.seed(123)
svm_model <- train(
  chd ~ .,
  data = train_data,
  method = "svmRadial",
  preProcess = c("center", "scale"),
  tuneLength = 10,
  trControl = ctrl,
  metric = "ROC"
)

cat("\nSVM Radial model:\n")

##
## SVM Radial model:

print(svm_model)

## Support Vector Machines with Radial Basis Function Kernel
##
## 324 samples
## 9 predictor
## 2 classes: 'No', 'Yes'
##
## Pre-processing: centered (9), scaled (9)
## Resampling: Cross-Validated (5 fold)
## Summary of sample sizes: 260, 259, 259, 260, 258
## Resampling results across tuning parameters:
##
##  C          ROC          Sens          Spec
##  0.25  0.7614452  0.8491694  0.47193676
##  0.50  0.7574843  0.8679956  0.39169960
##  1.00  0.7474766  0.8677741  0.38300395
##  2.00  0.7353258  0.8959025  0.33873518
##  4.00  0.7211887  0.9147287  0.30355731
##  8.00  0.7005390  0.9054264  0.22252964
## 16.00  0.6802816  0.9480620  0.13359684
## 32.00  0.6479644  0.9148394  0.19644269
## 64.00  0.6361413  0.9622370  0.08932806
## 128.00 0.6382644  0.9575858  0.09802372
##
## Tuning parameter 'sigma' was held constant at a value of 0.08260223
## ROC was used to select the optimal model using the largest value.
## The final values used for the model were sigma = 0.08260223 and C = 0.25.

# -----
# 13. Evaluation function
# -----
evaluate_model <- function(model, test_data, model_name) {
  pred_class <- predict(model, newdata = test_data)
  pred_prob <- predict(model, newdata = test_data, type = "prob")[, "Yes"]

  cm <- confusionMatrix(pred_class, test_data$chd, positive = "Yes")
  acc <- as.numeric(cm$overall["Accuracy"])

  roc_obj <- roc(response = test_data$chd, predictor = pred_prob, levels = c("No", "Yes"))
}
```

```

)
auc_val <- as.numeric(auc(roc_obj))

data.frame(
  Model = model_name,
  Accuracy = acc,
  AUC = auc_val
)
}

# -----
# 14. Compare models
# -----
results <- dplyr::bind_rows(
  data.frame(
    Model = "Ridge Logistic Regression",
    Accuracy = ridge_acc,
    AUC = ridge_auc
  ),
  evaluate_model(knn_model, test_data, "KNN"),
  evaluate_model(lda_model, test_data, "LDA"),
  evaluate_model(tree_model, test_data, "Decision Tree"),
  evaluate_model(rf_model, test_data, "Random Forest"),
  evaluate_model(svm_model, test_data, "SVM (Radial)")
)

## Setting direction: controls < cases
## Setting direction: controls < cases
## Setting direction: controls < cases
## Setting direction: controls < cases
## Setting direction: controls < cases

results <- results %>%
  dplyr::arrange(dplyr::desc(Accuracy))

cat("\nModel comparison results:\n")

##
## Model comparison results:

print(results)

##
##          Model Accuracy      AUC
## 1          KNN 0.7246377 0.7096065
## 2 Random Forest 0.7173913 0.7109954
## 3 SVM (Radial) 0.7173913 0.7259259
## 4 Ridge Logistic Regression 0.7028986 0.7486111
## 5          LDA 0.7028986 0.7495370
## 6 Decision Tree 0.6956522 0.6054398

# -----
# 15. Save outputs
# -----
write.csv(results, "model_comparison_results.csv", row.names = FALSE)
write.csv(group_summary,

```